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diff-diff: Comprehensive Difference-in-Differences Causal Inference for Python

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Software

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Summary

`diff-diff` is a Python library for Difference-in-Differences (DiD) causal inference analysis. It provides 20 estimators covering the full modern DiD toolkit - from classic two-group/two-period designs through heterogeneity-robust staggered adoption methods, synthetic control hybrids, and sensitivity analysis - under a consistent scikit-learn-style API. Most estimators accept an optional `SurveyDesign` object for design-based variance estimation with complex survey data, a capability absent from existing DiD software in any language; the underlying design-based variance methodology is derived in the companion preprint (Gerber, 2026). Point estimates are validated against established R packages to machine precision, with standard errors matching exactly or to sub-percent relative differences.

Statement of Need

Difference-in-differences is the most widely used quasi-experimental research design in applied economics and the social sciences. Since 2018, a wave of methodological advances has addressed fundamental limitations of the conventional two-way fixed effects (TWFE) estimator under staggered treatment adoption and heterogeneous effects (Roth et al., 2023). These modern methods - including Callaway and Sant'Anna (2021), Sun and Abraham (2021), Borusyak, Jaravel, and Spiess (2024), and others - are now standard practice in applied work.

These methods are well served in R and Stata, but Python lacks a unified DiD library. Practitioners working in Python-based data science workflows - increasingly common in industry settings for marketing measurement, product experimentation, and policy evaluation - must either context-switch to another language, reimplement methods from scratch, or rely on partial implementations scattered across unrelated packages.

`diff-diff` fills this gap by providing a single-import library that covers 20 estimators with a consistent API, survey-weighted inference, and numerical validation against R. It is also the companion software for the design-based variance framework of Gerber (2026), which establishes design-consistent standard errors for modern DiD estimators under complex survey designs. It targets both applied researchers who need rigorous econometric methods and data science practitioners who need accessible causal inference tools integrated into Python workflows.

State of the Field

The R ecosystem provides mature implementations across several packages: `did` (Callaway & Sant'Anna, 2021), `fixest` (Bergé, 2018), `synthdid` (Arkhangelsky et al., 2021), and `HonestDiD` (Rambachan & Roth, 2023); Stata offers `csdid` and `didregress`. Python coverage is partial and fragmented. `pyfixest` (The PyFixest Authors, 2025) brings `fixest`-style high-dimensional fixed-effects regression to Python, including Sun-Abraham, two-stage, and local-projections estimators, but is organized around its regression engine rather than the wider DiD taxonomy; `differences` implements Callaway-Sant'Anna group-time estimation; `CausalPy` offers Bayesian analysis of quasi-experiments, including synthetic control, without staggered-adoption support. General-purpose causal inference toolkits such as `DoWhy` and `EconML` target other identification strategies.

`diff-diff` was built as a new library, rather than as contributions to these packages, because its central contribution is cross-cutting: one estimator contract, one shared inference core, and an influence-function architecture that composes design-based survey variance across the estimator taxonomy, with per-estimator support documented in a compatibility matrix and unsupported combinations failing closed. To our knowledge, no existing DiD software in any language provides design-based variance estimation for complex survey data, and no Python package covers the modern estimator taxonomy end-to-end; `diff-diff` provides both, validated against the R reference implementations where they exist.

Key Features

Breadth of methods. `diff-diff` implements 20 estimators organized across the modern DiD taxonomy. Classic designs include two-group/two-period DiD, two-way fixed effects, and event-study estimation with period-specific effects. Heterogeneity-robust staggered-adoption estimators include Callaway-Sant'Anna (Callaway & Sant'Anna, 2021), Sun-Abraham (Sun & Abraham, 2021), imputation (Borusyak et al., 2024), two-stage (Gardner, 2022), stacked (Wing et al., 2024), efficient (Chen et al., 2025), and local-projections (Dube et al., 2025) approaches, together with reversible-treatment DiD for non-absorbing interventions (de Chaisemartin & D'Haultfœuille, 2020) and a ring-indicator estimator for spatial spillovers (Butts, 2023). Synthetic-control hybrids include synthetic DiD (Arkhangelsky et al., 2021) and the classic synthetic control method (Abadie et al., 2010). Extended designs include triple-difference and staggered triple-difference estimators (Ortiz-Villavicencio & Sant'Anna, 2025), continuous-treatment DiD with dose-response curves (Callaway et al., 2024), heterogeneous-adoption designs where no unit remains untreated (de Chaisemartin et al., 2026), nonlinear ETWFE (Wooldridge, 2023, 2025), and triply robust panel estimation (Athey et al., 2025). Separate diagnostic and sensitivity tools - outside the 20 estimators - include Goodman-Bacon decomposition (Goodman-Bacon, 2021), Honest DiD sensitivity analysis (Rambachan & Roth, 2023), placebo tests, and pre-trends power analysis (Roth, 2022).

Survey-weighted inference. A `SurveyDesign` class supports stratification, primary sampling units, finite population corrections, and probability weights. Variance estimation includes Taylor series linearization, five replicate weight methods (BRR, Fay's BRR, JK1, JKn, SDR), and survey-aware bootstrap. The design-based variance result - that the influence functions of modern DiD estimators satisfy the smoothness conditions of Binder (1983), so stratified-cluster linearization yields design-consistent standard errors - is derived in Gerber (2026). No other DiD package in any language provides integrated survey support.

Practitioner tooling. Beyond estimation, `diff-diff` includes a practitioner decision tree for estimator selection, an 8-step diagnostic workflow based on Baker et al. (2025), AI agent integration with structured next-steps guidance, and microdata aggregation utilities for converting individual-level survey responses into geographic-period panels suitable for DiD analysis.

Software Design

Every estimator implements a common contract: a scikit-learn-style `fit()` with `get_params()`/`set_params()` for configuration and rich results dataclasses with `summary()`, `to_dict()`, and `to_dataframe()`; the classic regression estimators additionally accept R-style formulas. Numerical work is deliberately centralized: estimators solve their least-squares problems through a single shared linear-algebra core, and analytical robust, cluster-robust, and survey variances route through one shared sandwich-estimator path, so numerical hardening - rank-deficiency guards, degrees-of-freedom corrections, small-cluster behavior - lands in one place. Estimators whose inference is inherently resampling-based - synthetic DiD's placebo and jackknife variance, for example - use method-specific variance paths. All estimators share one joint-inference contract: inference fields (standard error, t-statistic, p-value, confidence interval) are always computed together and become NaN together when inference is not identified, rather than silently reporting partial results.

Two design choices carry the survey capability and the deployment story. First, the regression- and influence-function-based estimators compute influence functions for their target parameters, so design-based variance - Taylor series linearization over strata and clusters, and replicate weights - routes through one shared survey-variance core rather than requiring per-estimator derivations; resampling-based estimators such as synthetic DiD and TROP compose survey designs through documented method-specific bootstrap, placebo, and jackknife paths. Supported design-estimator combinations are listed in a per-estimator compatibility matrix, and unsupported ones are rejected explicitly rather than silently approximated. Second, the runtime dependency footprint is minimal by policy - numpy, pandas, and scipy only - keeping the library easy to install in restricted industry environments; high-dimensional fixed effects are handled by within-transformation rather than by delegating to a heavier econometrics stack. An optional Rust backend (via PyO3) accelerates compute-intensive kernels such as synthetic-control weight solving and fixed-effects absorption; the Python implementation remains canonical, equivalence between backends is enforced by the test suite, and the library falls back to pure Python automatically when the extension is unavailable.

Research Impact Statement

`diff-diff` is the companion software of the design-based variance preprint (Gerber, 2026): the framework derived there is implemented here, and the preprint's numerical results are produced with the library. Correctness evidence ships with the repository as reproducible material. Golden-file benchmarks pin point estimates against R's `did`, `synthdid`, and `fixest` to machine precision (differences $<$ 1e-10), including the canonical MPDTA minimum-wage application of Callaway and Sant'Anna (2021), with standard errors matching exactly for core estimators such as Callaway-Sant'Anna and basic DiD. Survey variance is validated against R's `survey` package (Lumley, 2004) on three real complex-survey datasets - NHANES, RECS 2020, and the California API school data - with test gaps below 1e-8 and typically below 1e-10. The library is distributed on PyPI with tagged releases, has six months of continuous public development history (3,000+ commits), and is exercised by a CI test suite of more than 7,600 tests; 26 tutorial notebooks and full API documentation are published on Read the Docs, and machine-readable guides bundled in the wheel (`llms.txt`) make the library directly usable by AI-assisted analysis workflows.

AI Usage Disclosure

Generative AI tools were used in developing this software and manuscript. Anthropic's Claude models (the Opus, Sonnet, and Fable model families, via the Claude Code CLI) assisted with code generation and refactoring, test scaffolding, documentation, and drafting and editing of this manuscript. The author reviewed, modified, and validated all AI-generated code and text and made all primary architectural and methodological decisions. Numerical results were independently verified against established R reference packages (`did`, `synthdid`, `fixest`, `survey`) for every estimator with an R equivalent, and against the author's reference derivations or simulation otherwise. The author takes full responsibility for the accuracy and integrity of the software and this paper.

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